Akoizo Original Practice - Form 4 - Answer key

Division B | Roller Coaster | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) An elevator has IMA 0.5. If the counterweight drops 6 cm, how far can the ball rise?

A. 6 cm

B. 0.5 cm

C. 12 cm

D. 3 cm

Correct: C

2. (2 points) What happens if a timing section’s slope is too gentle?

Unit: Energy, friction, and the rolling ball.
Worked-case data: A solid ball drops 0.40 m from rest without losses. Find its speed.
Separate investigation observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

A. Nothing changes

B. The loop height increases

C. The ball may stop, ending timing

D. The ball speeds up

Correct: C

3. (2 points) Which action is NOT allowed during a scorable run?

Unit: Testing and competition day.
Worked-case data: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?
Separate investigation observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

A. Touching the device

B. Removing the pencil on “Go”

C. Watching the ball

D. Standing near the device

Correct: A

4. (2 points) With a 45 s Target Time, which run time earns the most Time Score?

A. 45.5 s

B. 44.9 s

C. 46.2 s

D. 43.0 s

Correct: A

5. (4 points) Explain why the ball must have a minimum speed at the top.

Unit: Loops: physics and scoring.
Worked-case data: A loop has a 9.0 cm path radius. What is the ideal minimum release height above the loop bottom, and a practical target?
Separate investigation observations:
Release 24 cm: Ball fell from top of loop in 6 of 10 trials.
Release 28 cm: Ball completed loop 10 of 10.
Loop path radius: 9 cm.

answer: Gravity alone pulls it down with g; it must move fast enough that its path needs at least that much centripetal acceleration. (4 points).

6. (4 points) What happens to scoring if the ball stops before the Finish Line?

answer: The Time Score is zero, but successful elevators and the loop before timing stopped still earn points. (4 points).

7. (4 points) What should a safety stop do?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

answer: Catch and hold the ball after it crosses the Finish Line, without it bouncing out. (4 points).

8. (4 points) Why must elevators be tested many times?

Unit: Elevators: levers, wheels, and pulleys.
Worked-case data: A second-class lever has IMA 1.6. It drives a movable pulley inverted to give IMA 0.5. Is the combination legal, and how far does the ball rise if the effort moves 8 cm?
Separate investigation observations:
Elevator test: Counterweight drops 5 cm; ball rises 9 cm.
Requirement: Ball must rise at least 10 cm.
Team claim: Elevator earns 50 points.

answer: A stalled elevator stops timing, can lose the Time Score, and earns no elevator points. (4 points).

9. (4 points) Why might a steel ball be more consistent than a plastic one?

Unit: Energy, friction, and the rolling ball.
Worked-case data: A solid ball drops 0.40 m from rest without losses. Find its speed.
Separate investigation observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

answer: Its greater mass makes it less affected by small bumps, friction, and air resistance. (4 points).

10. (6 points) Solve this case and explain your reasoning: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?

reasoning: Aim for about 52.5 s, half a second past the target: about 1.9 s more.
1.9 s ÷ 0.4 s/turn about 5 turns, giving about 52.6 s.
Make 5 turns, then declare the scorable run. (3 points).

conclusion: Add 5 turns for about 52.6 s. (3 points).

11. (6 points) Solve this case and explain your reasoning: Calibration: base path 34 s; each slope-screw turn adds 0.8 s; removable spiral adds 9 s. Plan for a Target Time of 46 s.

reasoning: Add the spiral: 34 + 9 = 43 s.
Need about 3 s more: 3 ÷ 0.8 about 4 screw turns, giving 46.2 s.
Use 4 turns (46.2 s), inside the 46.0-46.99 s window that earns full points. (3 points).

conclusion: Spiral plus 4 screw turns gives about 46.2 s. (3 points).

12. (2 points) Define “Loop height” in the context of Roller Coaster.

answer: Vertical distance from POI to highest inside point. (2 points).

13. (2 points) Define “Run Score” in the context of Roller Coaster.

answer: Size + Time + Elevator + Loop. (2 points).

14. (2 points) Define “Footprint” in the context of Roller Coaster.

answer: Area the device occupies on the table. (2 points).

15. (2 points) Define “IMA < 1” in the context of Roller Coaster.

answer: Load moves farther than the effort. (2 points).

16. (2 points) Define “Rolling resistance” in the context of Roller Coaster.

answer: Energy loss from deformation as the ball rolls. (2 points).

17. (2 points) Define “Calibration table” in the context of Roller Coaster.

answer: Record of time added by each adjustment. (2 points).

18. (2 points) Define “Slope adjuster” in the context of Roller Coaster.

answer: Device to fine-tune a track section’s tilt. (2 points).